

WHATNEXT VISION MOTORS

Salesforce CRM Implementation Project Report

Project Title: WhatNext Vision Motors – Salesforce CRM System

Project Type: Team Project

Developed By:

- Praveen Alexander
- Lokeshwar C
- Mira Asmitha S
- Venkateswaran A
- Muthukumaran R

Course: Salesforce

Institution: Tagore Engineering College

Guide: Asst. Professor Suganthi J

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ABSTRACT

WhatNext Vision Motors is an innovative organization in the automotive industry that focuses on delivering advanced and customer-centric mobility solutions. With increasing demand for digital transformation, the company aims to improve its operational efficiency and customer engagement through technology.

This project implements a Salesforce CRM system that automates key business processes such as vehicle ordering, dealer assignment, stock validation, and customer communication. By leveraging Salesforce Flows and Apex programming, the system ensures accuracy, reduces manual intervention, and provides a scalable solution for future growth.

1. INTRODUCTION

In today's competitive automotive market, efficient customer relationship management is essential for business success. WhatNext Vision Motors seeks to modernize its sales and service operations using Salesforce CRM.

The system is designed to improve the overall customer journey by automating repetitive tasks and ensuring seamless coordination between customers, dealers, and internal teams.

The key focus areas of the project include:

- Enhancing the vehicle ordering experience
 - Automating dealer assignment based on location
 - Efficient stock management
 - Improving communication through automated notifications
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2. PROBLEM STATEMENT

Traditional vehicle ordering systems are often manual and inefficient, leading to delays and errors. Customers may unknowingly place orders for vehicles that are out of stock, and dealer assignment is typically handled manually, which increases processing time.

Additionally, there is limited communication between the system and customers, resulting in poor user experience and lack of transparency. Tracking customer activities and managing service requests also become challenging without automation.

This project addresses these issues by introducing a smart and automated CRM solution using Salesforce technologies.

3. PROJECT OBJECTIVES

The primary objective of this project is to design and implement an automated CRM system that improves operational efficiency and customer satisfaction.

The specific objectives are:

- Automate the vehicle order processing system
- Assign the nearest dealer automatically based on customer location
- Prevent orders when stock is unavailable
- Automate reminders for scheduled test drives

- Reduce manual effort and errors
 - Enhance overall customer experience
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4. SYSTEM OVERVIEW

The proposed system is developed on the Salesforce Lightning Platform, which provides a robust and scalable environment for CRM solutions. It integrates multiple components such as custom objects, automation tools, and Apex programming.

The system utilizes the following technologies:

- Salesforce Lightning Platform for application development
 - Custom Objects for structured data storage
 - Record-Triggered Flows for automation
 - Apex Triggers for business logic implementation
 - Batch Apex and Scheduler for background processing
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5. SYSTEM ARCHITECTURE

The system architecture defines the relationships between different entities involved in the CRM system. It ensures smooth interaction between customers, vehicles, dealers, and service processes.

In this model, a customer can place a vehicle order, which is linked to both the selected vehicle and an assigned dealer. Customers can also schedule test drives, while dealers handle service requests related to vehicles.

This interconnected structure enables efficient data flow and process automation across the system.

6. FUNCTIONAL REQUIREMENTS

The system provides various functional capabilities to support business operations. These requirements are categorized into CRM implementation, automation, Apex logic, and batch processing.

6.1 Salesforce CRM Implementation

The CRM system is responsible for storing and managing all business-related data. It maintains detailed records of vehicles, dealers, customers, and service requests.

Key functionalities include:

- Storing vehicle information such as model, price, and stock
- Maintaining dealer and customer records
- Tracking customer interactions
- Managing vehicle service requests

6.2 Process Automation

Automation plays a vital role in improving efficiency and reducing manual work. The system automatically assigns dealers and validates stock before processing orders.

Automation features include:

- Auto-assignment of the nearest dealer
- Prevention of orders for out-of-stock vehicles
- Automated email reminders for test drives

6.3 Apex and Triggers

Apex is used to implement custom business logic that cannot be handled by declarative tools alone. Triggers ensure that validations and updates occur at the right time.

Key functionalities include:

- Validating stock availability before order creation
- Automating dealer assignment logic
- Ensuring modular and reusable code design

6.4 Batch Jobs

Batch processing is used for handling large volumes of data efficiently. It runs periodically to update system records and maintain data accuracy.

Batch job functionalities include:

- Periodic stock verification
- Updating vehicle availability status
- Sending notifications when required

7. DATA MODEL

The data model defines the structure of the system and the relationships between different objects. It ensures efficient data storage and retrieval.

The main objects created in the system are:

- Vehicle
- Vehicle Dealer
- Vehicle Customer
- Vehicle Order
- Vehicle Test Drive
- Vehicle Service Request

Each object contains relevant fields. For example, the Vehicle object includes model, price, stock quantity, and status, while the Vehicle Order object includes customer details, dealer assignment, and order status.

8. SYSTEM DESIGN

The system design illustrates how different processes are executed within the application.

Order Process

The order process begins when a customer creates an order. The system then checks stock availability and assigns the nearest dealer. Once validated, the order is confirmed and processed successfully.

Test Drive Process

In the test drive process, a customer schedules a test drive. The system waits for a predefined time and automatically sends a reminder email, ensuring better customer engagement.

9. IMPLEMENTATION

The implementation phase involves configuring Salesforce and developing automation logic.

9.1 Salesforce Setup

Initially, a Salesforce Developer account was created. Custom objects were designed to represent business entities, and a Lightning application was built to provide a user-friendly interface.

9.2 Flow Automation

Salesforce Flows were used to automate key processes. Dealer assignment is performed based on customer location, while reminder flows ensure timely communication with customers.

10. APEX IMPLEMENTATION

Apex programming is used to handle complex logic such as stock validation and updates.

Vehicle Order Trigger

```
trigger VehicleOrderTrigger on Vehicle__Order__c (before insert, before update) {  
    for(Vehicle__Order__c order : Trigger.new){  
        Vehicle__c v = [SELECT Stock_Quantity__c FROM Vehicle__c WHERE Id =  
:order.Vehicle__c];  
  
        if(v.Stock_Quantity__c <= 0){  
            order.addError('Vehicle is out of stock');  
        }  
    }  
}
```

Trigger Handler

```
public class VehicleOrderHandler {  
    public static void updateStock(List<Vehicle__Order__c> orders){  
        Set<Id> ids = new Set<Id>();  
  
        for(Vehicle__Order__c o : orders){  
            if(o.Status__c == 'Confirmed'){  
                ids.add(o.Vehicle__c);  
            }  
        }  
    }  
}
```

```

    }

    List<Vehicle__c> vehicles = [SELECT Id, Stock_Quantity__c FROM
Vehicle__c WHERE Id IN :ids];

    for(Vehicle__c v : vehicles){
        v.Stock_Quantity__c -= 1;
    }

    update vehicles;
}
}

```

Batch Apex

```

global class VehicleBatch implements Database.Batchable<SObject> {

    global Database.QueryLocator start(Database.BatchableContext bc){
        return Database.getQueryLocator('SELECT Id, Stock_Quantity__c FROM
Vehicle__c');
    }

    global void execute(Database.BatchableContext bc, List<Vehicle__c> scope){
        for(Vehicle__c v : scope){
            if(v.Stock_Quantity__c == 0){
                v.Status__c = 'Out of Stock';
            }
        }
        update scope;
    }

    global void finish(Database.BatchableContext bc){}
}

```

Scheduler

```

global class VehicleScheduler implements Schedulable {
    global void execute(SchedulableContext sc){
        Database.executeBatch(new VehicleBatch(), 200);
    }
}

```

11. TESTING

The system was tested using multiple scenarios to ensure reliability and correctness. Orders placed with zero stock were blocked, confirming the validation logic. When an order was confirmed, the stock quantity was reduced accordingly. Test drive scheduling successfully triggered automated email reminders.

12. RESULTS

- Automated dealer assignment
 - Real-time stock validation
 - Improved operational efficiency
 - Reduced manual workload
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13. ADVANTAGES

The system offers several advantages, including faster processing of orders, reduction in human errors, improved customer satisfaction, and a scalable architecture that supports future enhancements.

14. WHAT WE LEARNED

Through this project, we gained practical knowledge in Salesforce development and CRM design. We learned about data modeling, object relationships, Lightning App Builder, Flow automation, Apex programming, and batch processing techniques.

15. FUTURE ENHANCEMENTS

- AI-based vehicle recommendations
 - Mobile application integration
 - SMS notification system
 - GPS-based dealer mapping
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16. CONCLUSION

This project demonstrates the effectiveness of Salesforce CRM in automating real-world business processes. It significantly improves efficiency, enhances customer satisfaction, and provides a scalable solution for future growth.

17. REFERENCE:

Demo Video Link :

https://drive.google.com/file/d/1m3bVoZSsf3OUuGMDS8xbAlV0F4bi6YjS/view?usp=drive_link

Salesforce Link :

Team Lead - <https://www.salesforce.com/trailblazer/fkx2j9usaa6tbrvxl>

Member 1 - <https://www.salesforce.com/trailblazer/b97vxayrefnus2j38i>

Member 2 - <http://salesforce.com/trailblazer/dnseykjgaidry7whkp>

Member 3 - <https://www.salesforce.com/trailblazer/the1kyepg3r3girplw>

Github Repo Link : <https://github.com/Alex001-debug/NM-2026>